

The Invisible Mechanism Behind Correct Sources Producing Wrong AI Answers

Interpretation Failure, Semantic Fidelity, and Reality Drift

Invisible Mechanism Note #1 — Reality Drift Framework

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The Feeling

The source is correct.

The passage is relevant.

The facts are present.

The answer is still wrong.

This is one of the most confusing failures in AI.

Nothing obvious appears broken.

Retrieval worked.

The model found the right material.

The response sounds coherent.

Yet the meaning has shifted.

The Invisible Mechanism

The core mechanism is simple:

factual fragments can survive while the relationships that give them meaning are lost.

AI systems do not only retrieve information.

They reconstruct an answer from compressed representations of that information.

During reconstruction, the system may preserve the facts while changing how those facts relate to one another.

The evidence remains.

The interpretation drifts.

How the Shift Happens

The sequence is:

correct source → compression → reconstruction → fluent answer → semantic mismatch

A source contains facts, context, sequence, emphasis, and relationships.

Retrieval selects only part of that structure.

Compression reduces it further.

The model then rebuilds an answer using learned patterns.

The final response may remain plausible while no longer preserving the source's intended meaning.

Why Retrieval Is Not Enough

Retrieval is often treated as the grounding layer.

If the system found the right document, the answer is assumed to be grounded.

But access to a source does not guarantee faithful interpretation.

The model can flatten distinctions.

It can merge separate claims.

It can reverse emphasis.

It can preserve wording while changing implication.

The failure occurs after retrieval but before meaning is stabilized.

Semantic Fidelity

Semantic fidelity is the preservation of meaning across representation, compression, retrieval, and reconstruction.

High semantic fidelity means the answer preserves the structure of the source.

Not only its facts.

Not only its vocabulary.

Its relationships, boundaries, qualifications, and intended claim.

A response can be factually accurate at the sentence level and still be semantically unfaithful at the level of the argument.

Reality Drift

Within the **Reality Drift framework**, this failure begins when the reconstructed answer becomes more influential than the source it was meant to represent.

The response sounds complete.

The source becomes invisible.

Users trust the generated interpretation because it is fluent, direct, and easier to consume.

The representation replaces the material it was supposed to preserve.

Reality Drift occurs when a correct source produces an answer that remains coherent while losing semantic contact with the source.

The Hidden Failure

The system does not need to invent facts.

It only needs to rearrange meaning.

Over time:

- retrieval replaces reading
- fragments replace structure
- fluency replaces fidelity
- plausibility replaces interpretation
- coherence replaces contact

The answer may be grounded in the source.

It may still fail to mean what the source meant.

The system retrieved the evidence and reconstructed the wrong meaning.

Factual correctness can survive after semantic fidelity has failed.

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